SENSOR

NASPIER V2
IMU BOARD

File: SENSOR_IMU.kicad_sch

HIROSE 34P
CONNECTOR

CORE

File: CORE.kicad_sch

CARRIER

PAB STANDARD
CARRIER BOARD

File: CARRIER.kicad_sch

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	NASPIER CORE: SYSTEM (CORE / SENSOR / CARRIER)
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version	Size / Page 1 Total 14
		REV0 VER0	A3 / VAP-SAN

HIROSE 34P
FFC CONNECTOR

CORE_H743

STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)

File: CORE_H743.kicad_sch

CORE_POWER

IMU BUSSES 3.3V
REGULATORS

FC & AUX 3.3V
REGULATORS

File: CORE_POWER.kicad_sch

CORE_ONBOARD

AT24C64D
EEPROM-I2C

FRAM
SPI

ICM-45686
IMU-SPI

MS5611
BARO-I2C

File: CORE_ONBOARD.kicad_sch

CORE_CONNECTORS

HIROSE 100P
CONNECTOR

HIROSE 50P
CONNECTOR

File: CORE_CONNECTORS.kicad_sch

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	CORE BOARD
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version	Size / Page 2 Total 14
		REV0 VER0	A4 / VAP-SAN

U1A
STM32H743IIK6

U1B
STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)
NON-STANDARD PIXHAWK CHIP

FMU 3V3 DIGITAL LINE CAPS

FMU 3V3 ANALOG LINE CAPS

R-G-B LEDS

CRYSTALS

SPI4 EXPANSION

REV SENS DO NOT PLACE

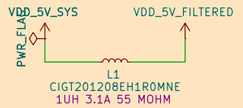
RTC BATTERY LINE

DEBUG INTERFACE

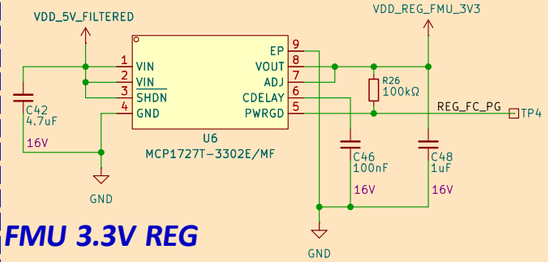
THIS CONNECTOR JUST FOR DEBUG/PROGRAMMING POINTS GROUPS

Schematic	Naspier Core	Create at	2026-09-20
Board	NASPIER_CORE_REV0_VER0	Update at	2026-10-15
Drawn	Ahmet Vapurcu	Page	CORE_H743
Reviewed			
	Naspier Core		
	Version: REV0 VER0	Size: A3	Page 3 Total 14
			VAP-SAN

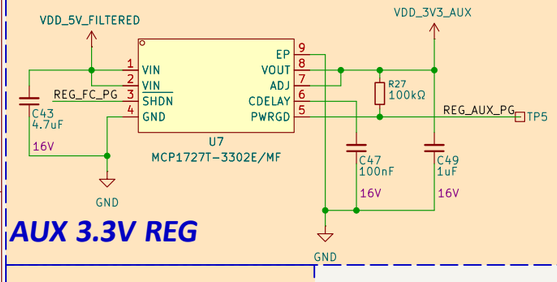
5V LINE FILTER



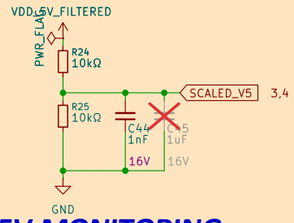
FMU 3.3V REG



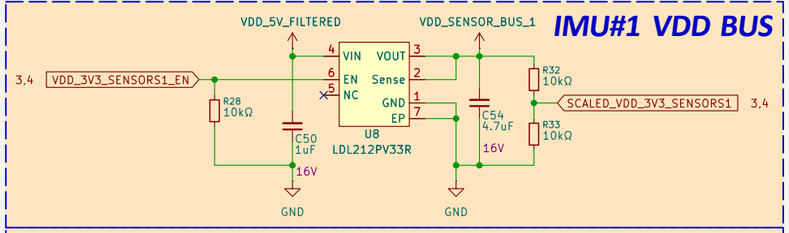
AUX 3.3V REG



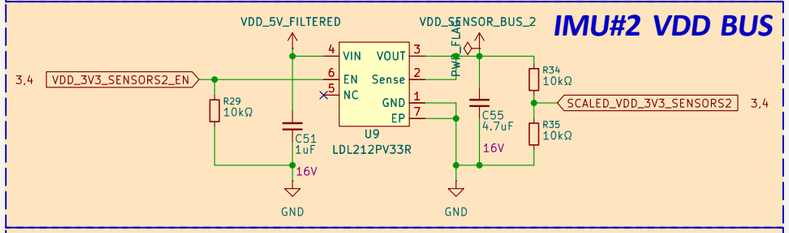
5V MONITORING



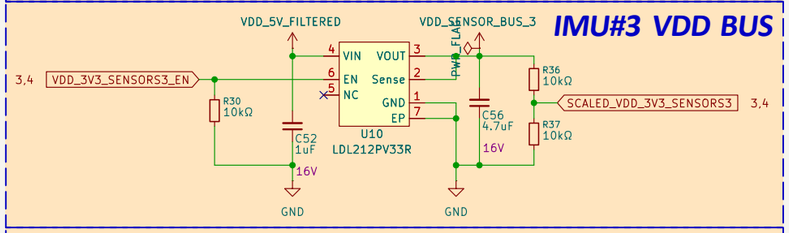
IMU#1 VDD BUS



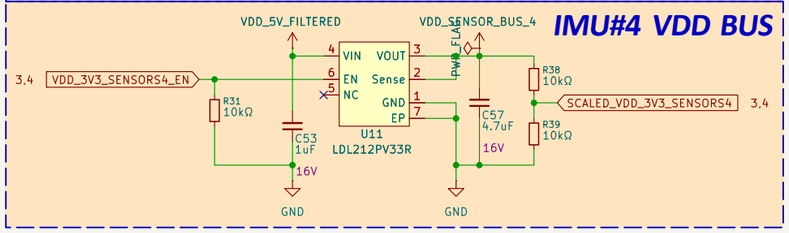
IMU#2 VDD BUS



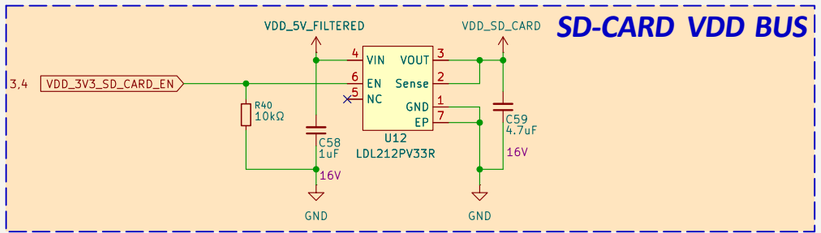
IMU#3 VDD BUS



IMU#4 VDD BUS

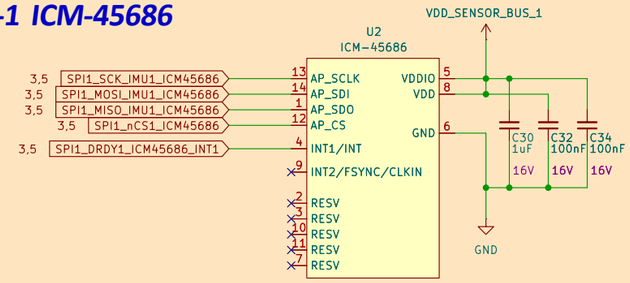


SD-CARD VDD BUS

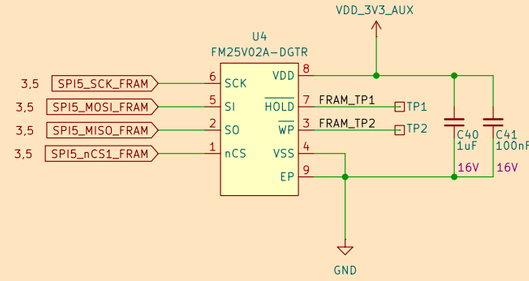


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CORE_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 4 Total 14
		REV0 VER0	A3	VAP-SAN

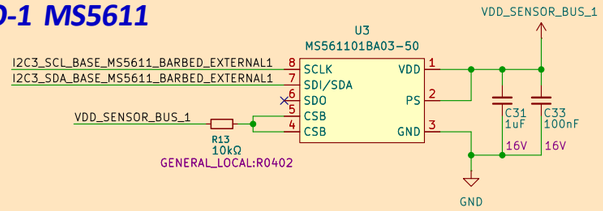
IMU-1 ICM-45686



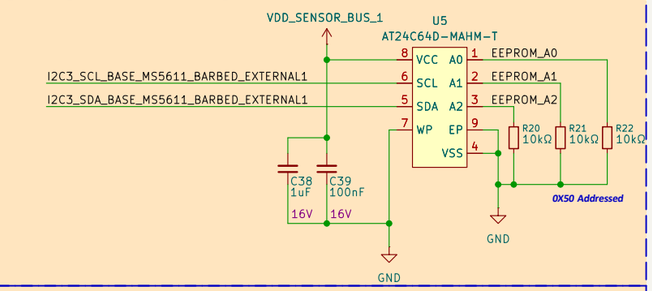
FRAM



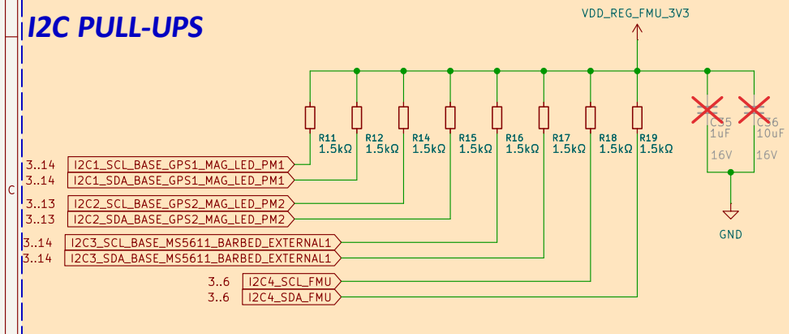
BARO-1 MS5611



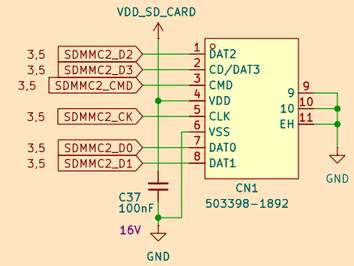
ONBOARD EEPROM



I2C PULL-UPS



SD-CARD

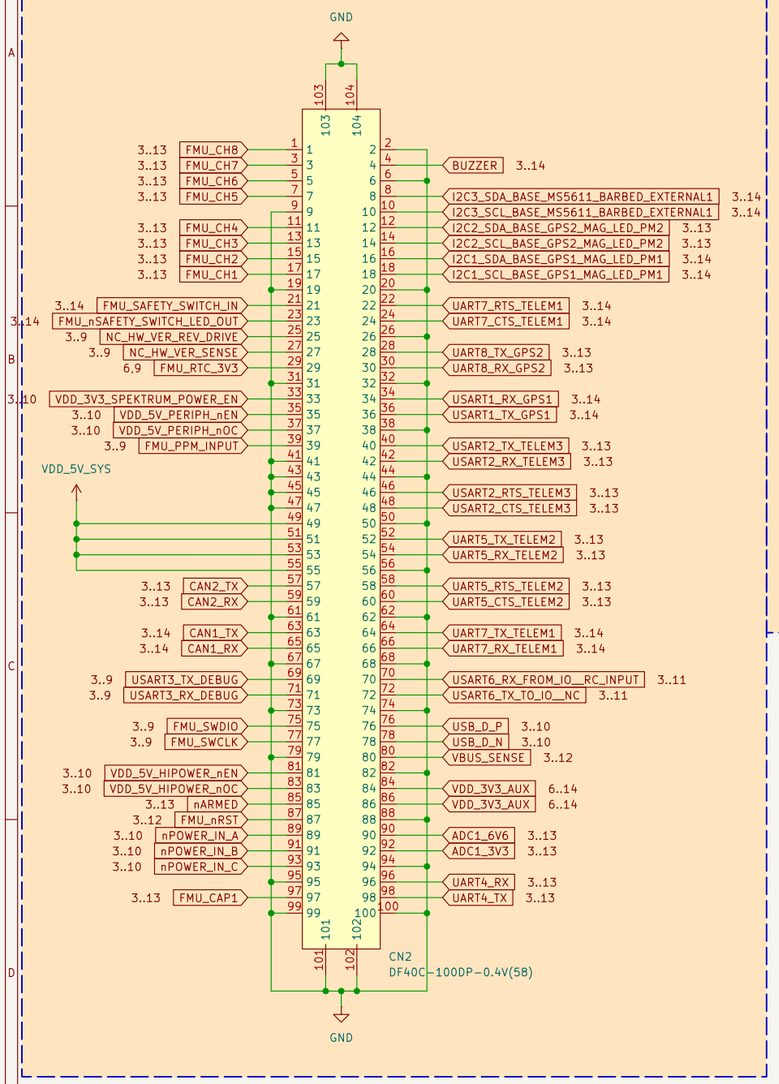


Push-In
Push-Out

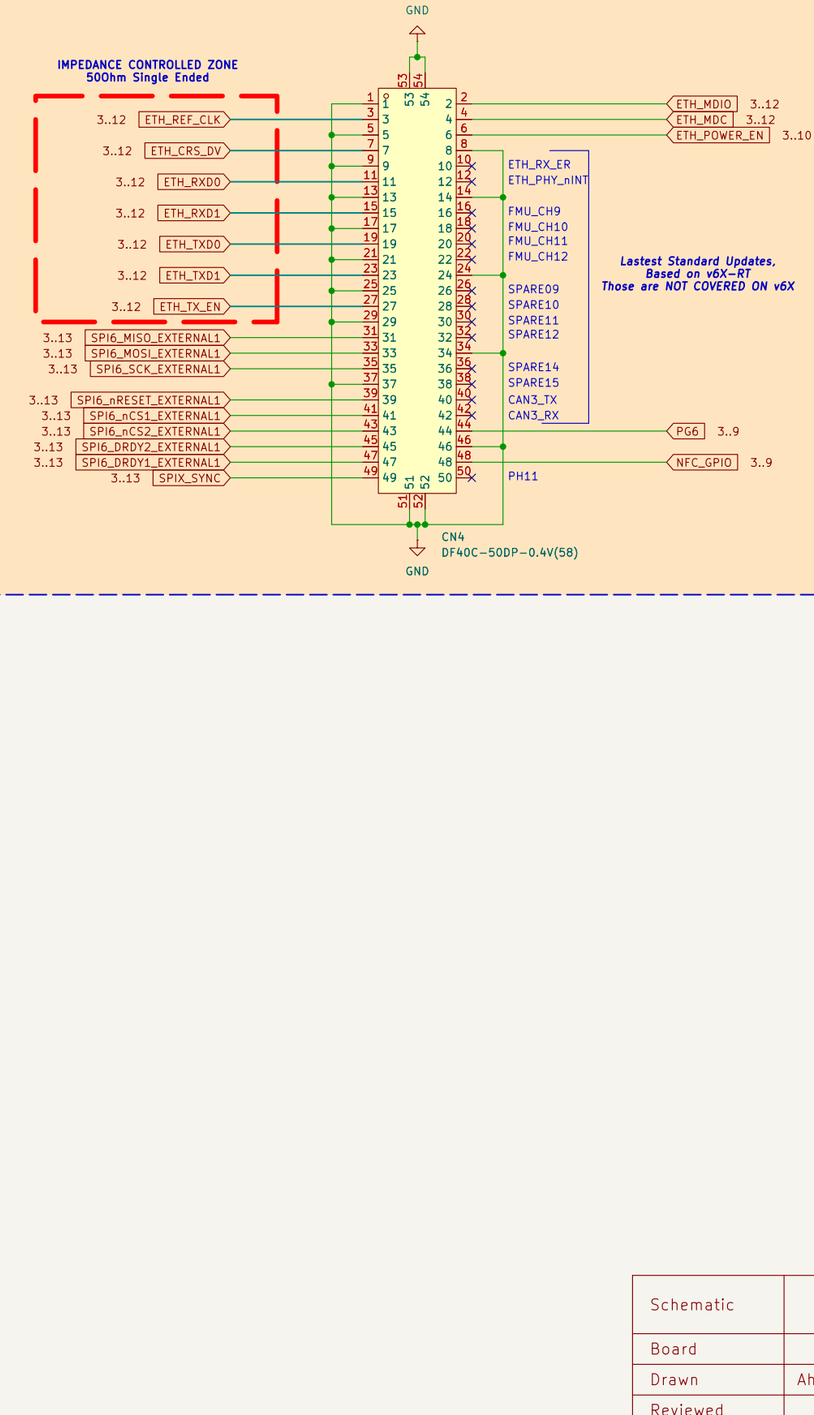
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CORE_ELEMENTS**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			

Version	Size	Page 5 Total 14
REV0 VER0	A3	VAP-SAN

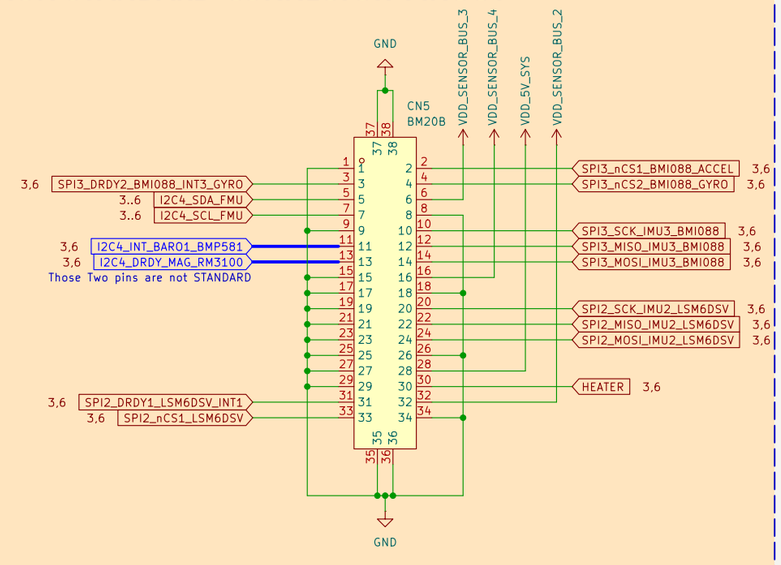
PAB X1 100P HIROSE



PAB X2 50P HIROSE

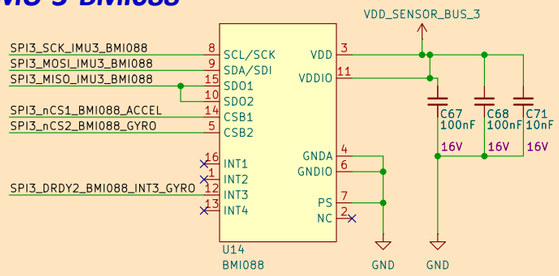


PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT

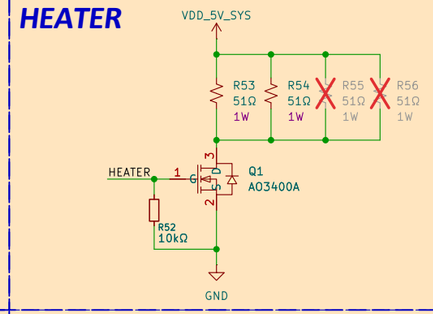


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CORE_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 6 Total 14
		REV0 VER0	A3	VAP-SAN

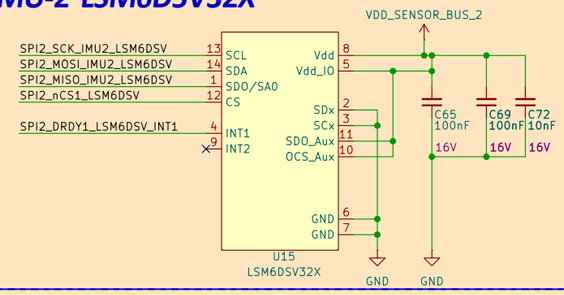
IMU-3 BMI088



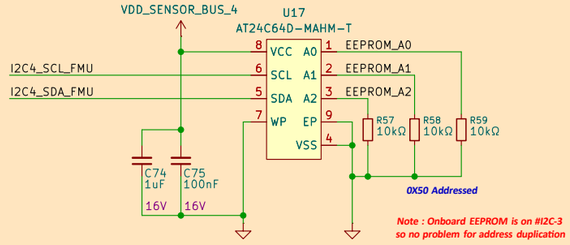
HEATER



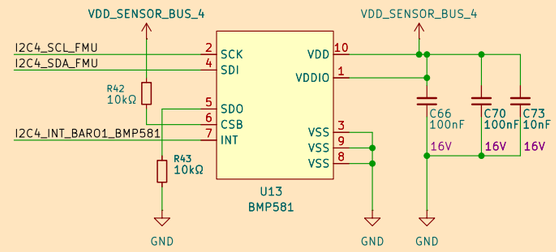
IMU-2 LSM6D5V32X



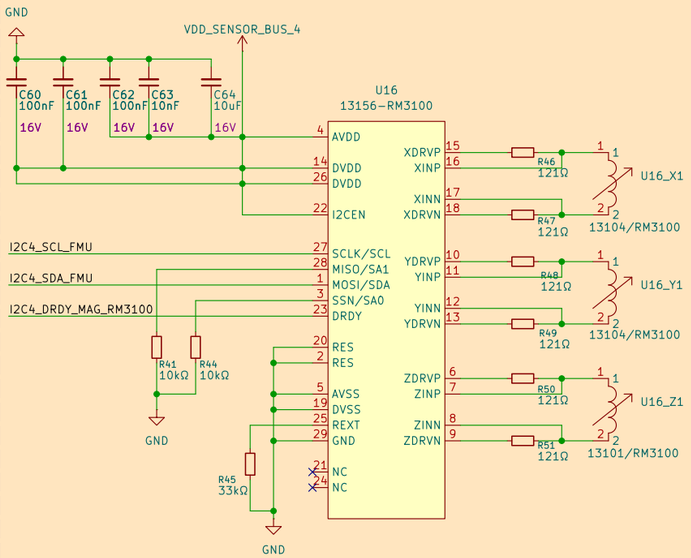
EEPROM



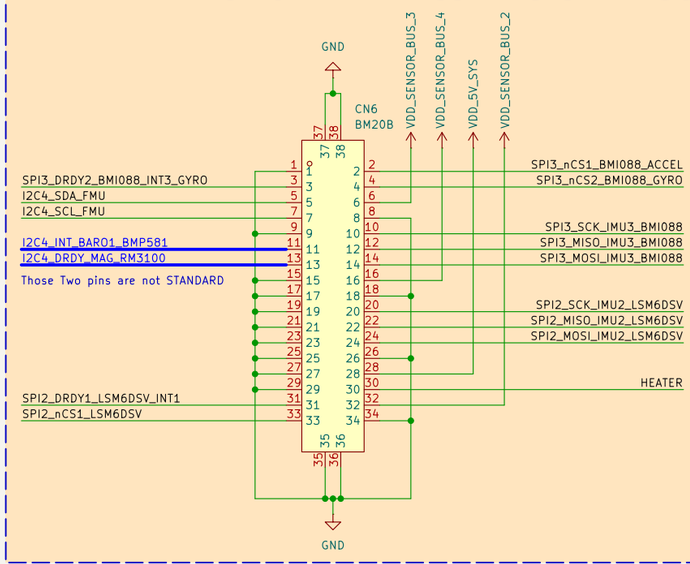
BARO-2 BMP581



MAG-RM3100



PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT



Schematic	**Naspier Core**	Create at	2026-09-20
Board	NASPIER_CORE_REV0_VER0	Update at	2026-10-15
Drawn	Ahmet Vapurcu	Page	**SENSOR_BOARD**
Reviewed		**Naspier Core**	
		Version: REV0 VER0	Size: A3
		Page 7 Total 14	VAP-SAN

CARRIER_B2B

File: CARRIER_B2B.kicad_sch

CARRIER_CONN_1

File: CARRIER_CONN_1.kicad_sch

CARRIER_IO_MCU

File: CARRIER_IO_MCU.kicad_sch

CARRIER_POWER

File: CARRIER_POWER.kicad_sch

CARRIER_CONN_2

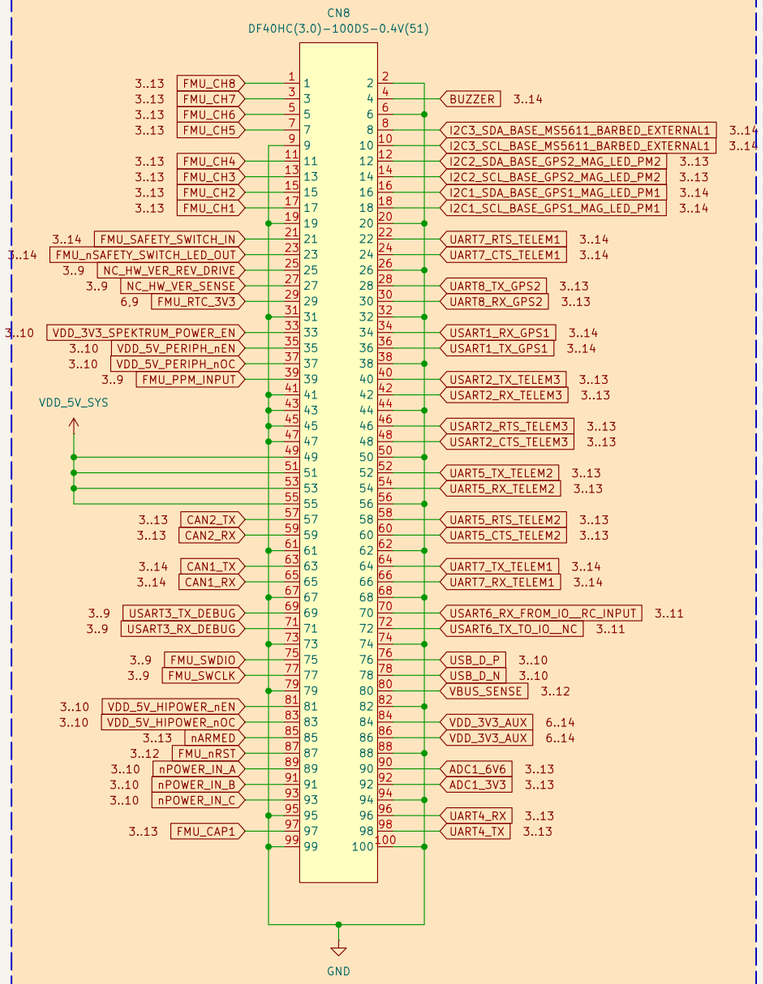
File: CARRIER_CONN_2.kicad_sch

CARRIER_ELEMENTS

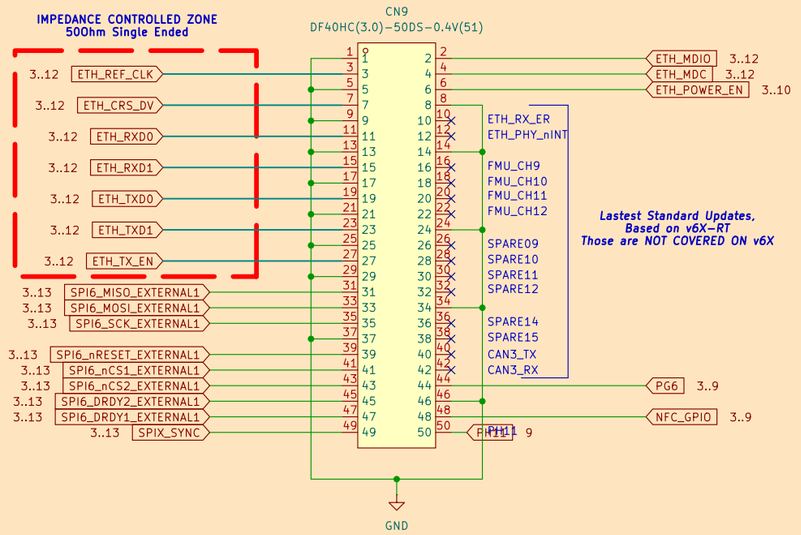
File: CARRIER_ELEMENTS.kicad_sch

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER BOARD**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 8 Total 14	
		REV0 VER0 / A4	VAP-SAN

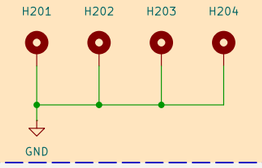
PAB X1 100P HIROSE



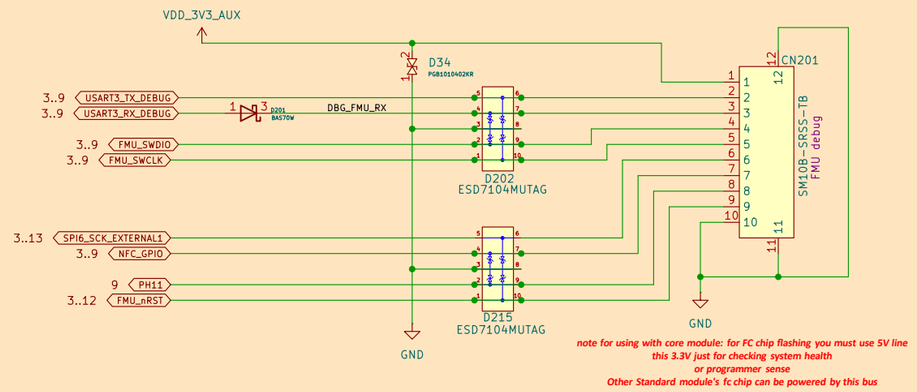
PAB X2 50P HIROSE



MountingHole M2 x4
CORE SCREWED STANDOFFS

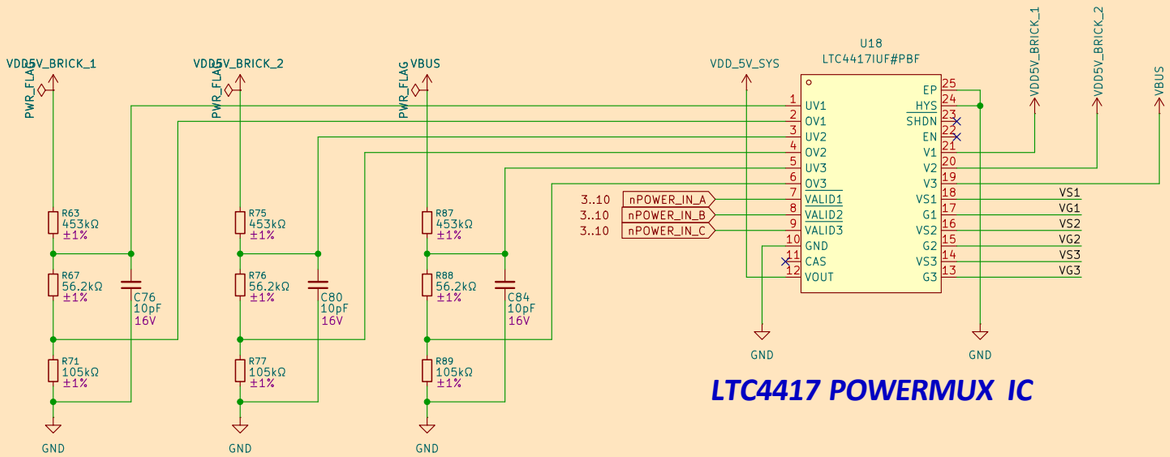


FMU DEBUG

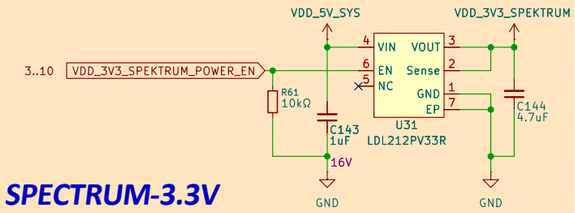


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 9 Total 14	
		REV0 VER0 / A3 / VAP-SAN	

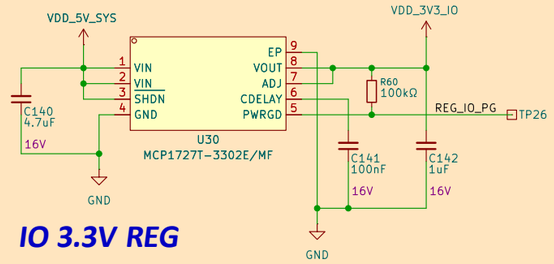
LTC4417 POWERMUX IC



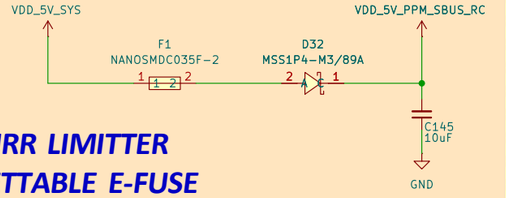
SPECTRUM-3.3V



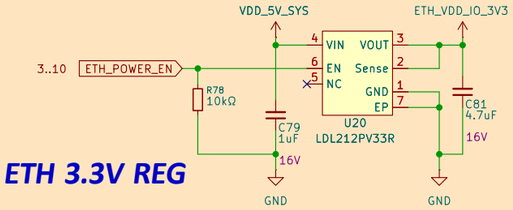
IO 3.3V REG



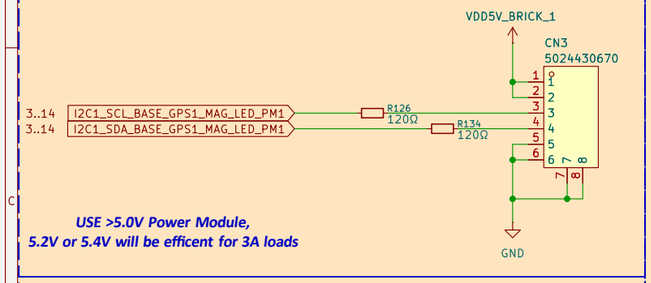
CURR LIMITTER RESETTABLE E-FUSE



ETH 3.3V REG

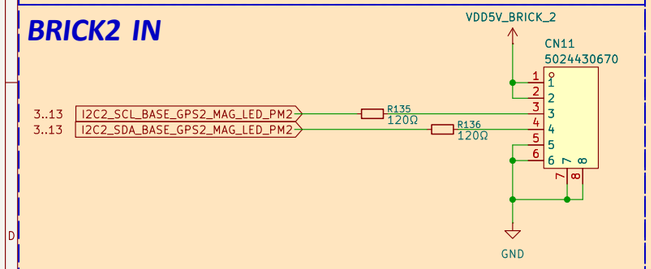


BRICK1 IN

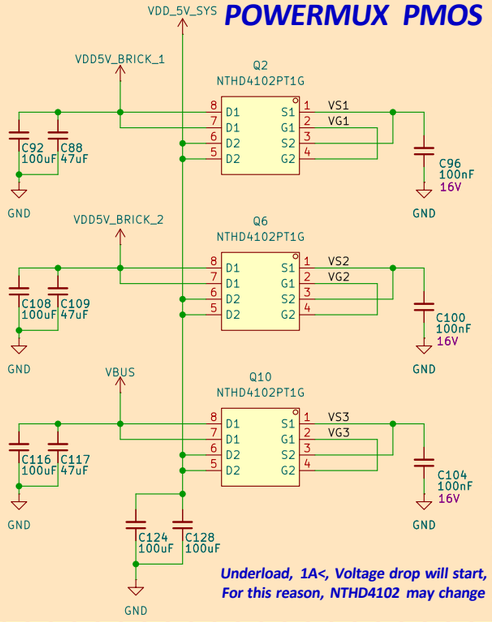


USE >5.0V Power Module,
5.2V or 5.4V will be efficent for 3A loads

BRICK2 IN

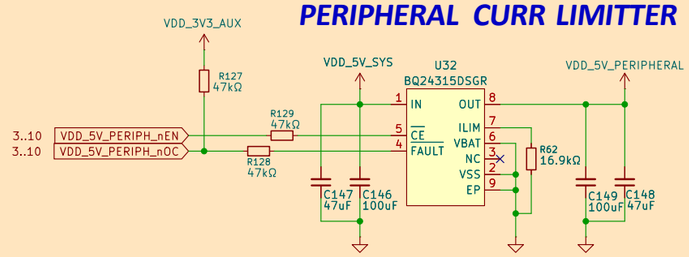


POWERMUX PMOS

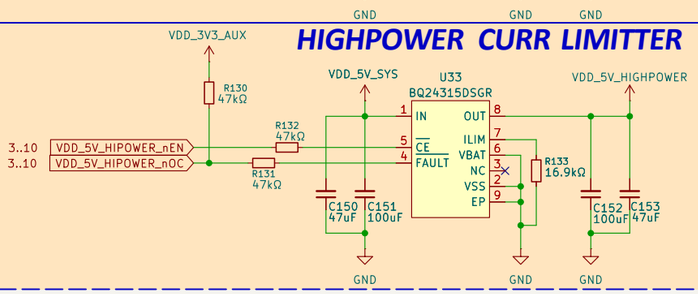


Underload, 1A<, Voltage drop will start,
For this reason, NTHD4102 may change

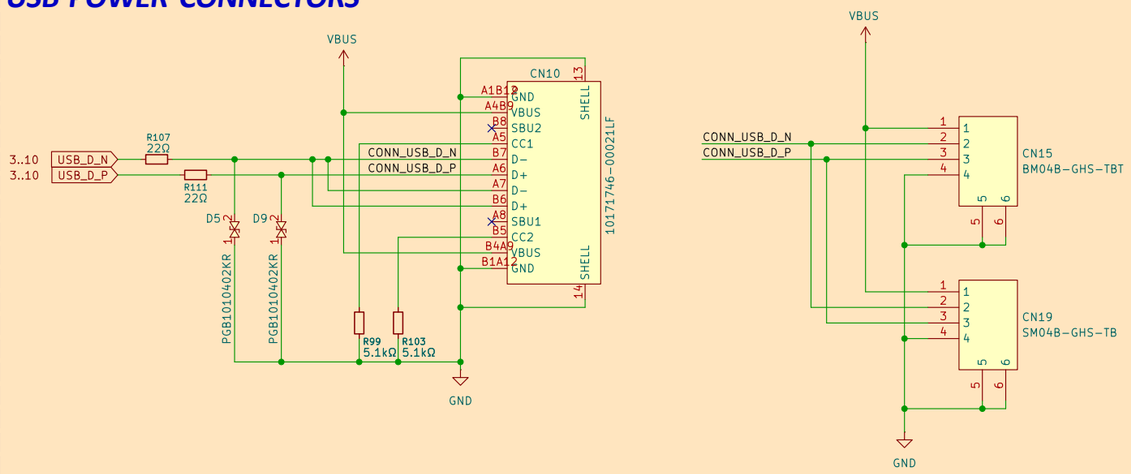
PERIPHERAL CURR LIMITTER



HIGHPOWER CURR LIMITTER

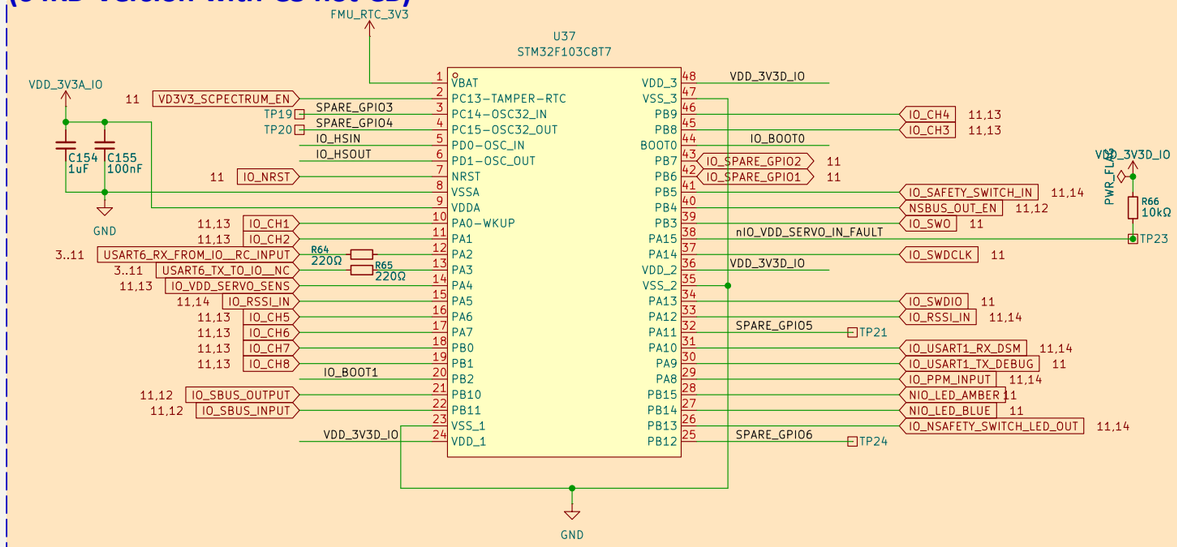


USB POWER CONNECTORS

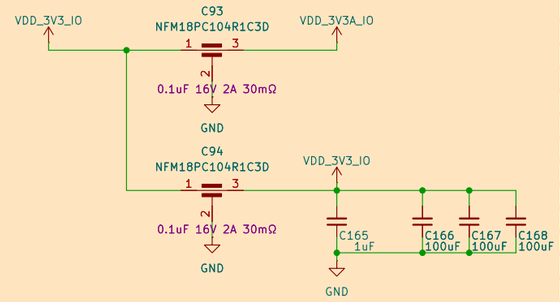


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 10 Total 14	
		REV0 VER0 / A3 / VAP-SAN	

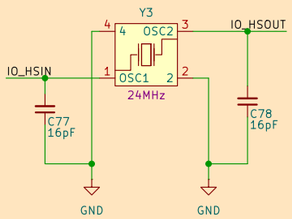
IO F103 CHIP
(64KB Version with C8 not CB)



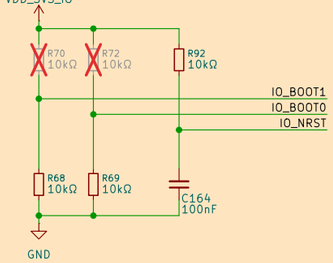
IO POWER FILTER



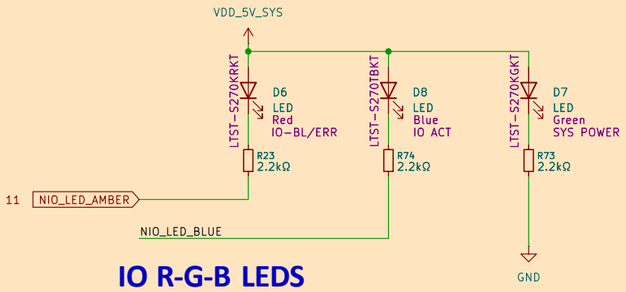
IO CRYSTAL



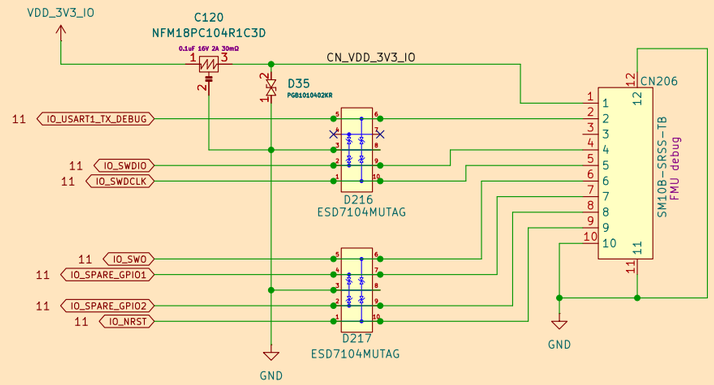
IO BOOT-NRST



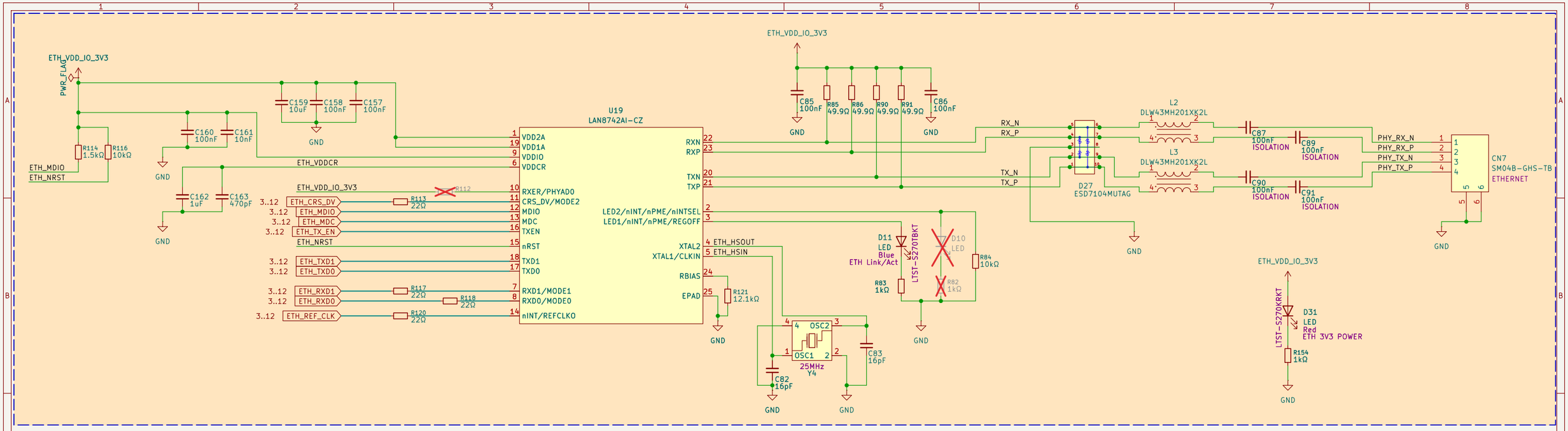
IO R-G-B LEDS



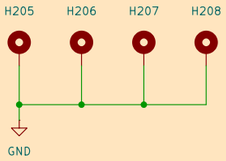
IO DEBUG



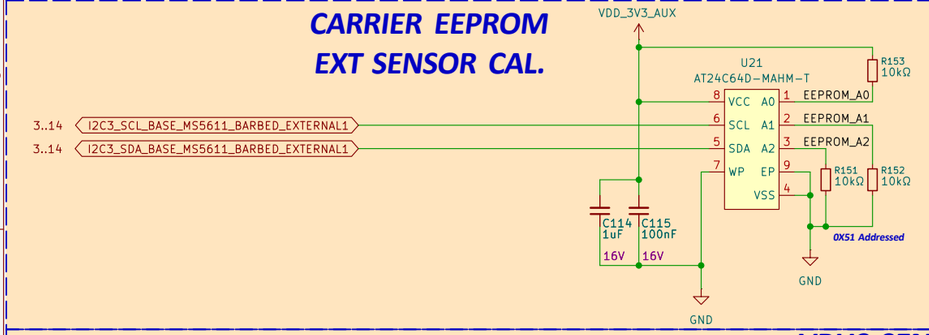
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_IO_MCU**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 11 Total 14
		REV0 VER0	A3	VAP-SAN



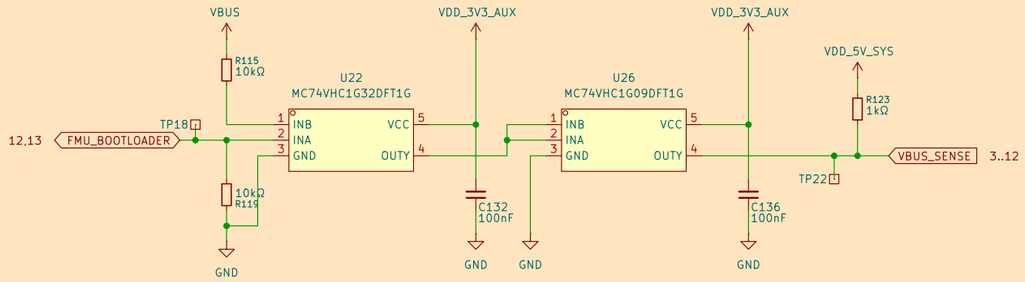
MountingHole M3 x4 BASEBOARD HOLES



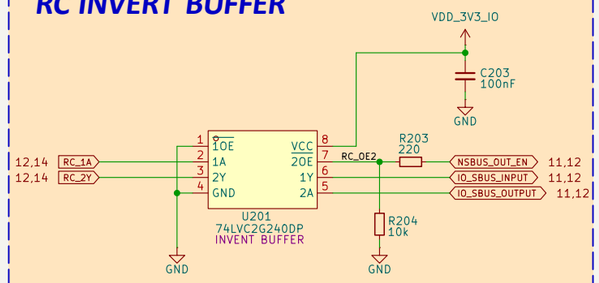
CARRIER EEPROM EXT SENSOR CAL.



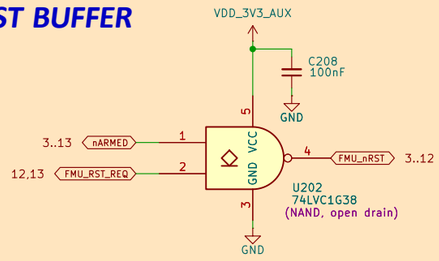
VBUS SENS CIRCUIT



RC INVERT BUFFER

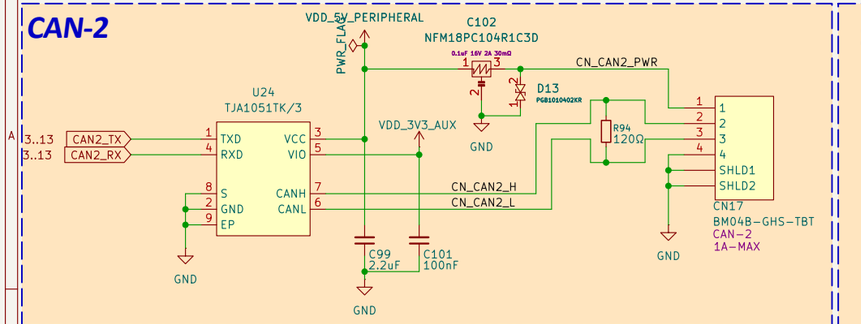


NRST BUFFER

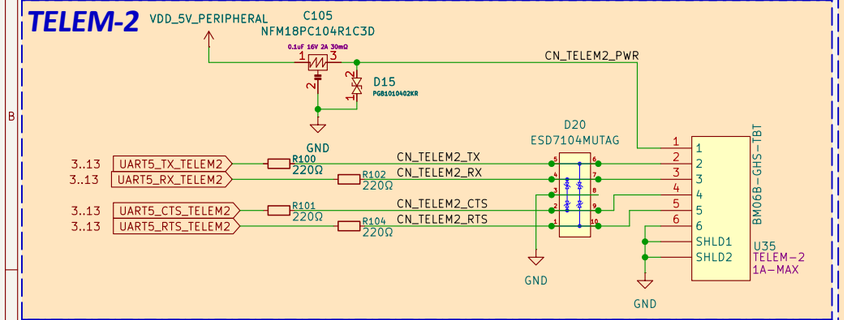


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	**CARRIER_ELEMENTS**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 12 Total 14	
		REV0 VERO / A3 / VAP-SAN	

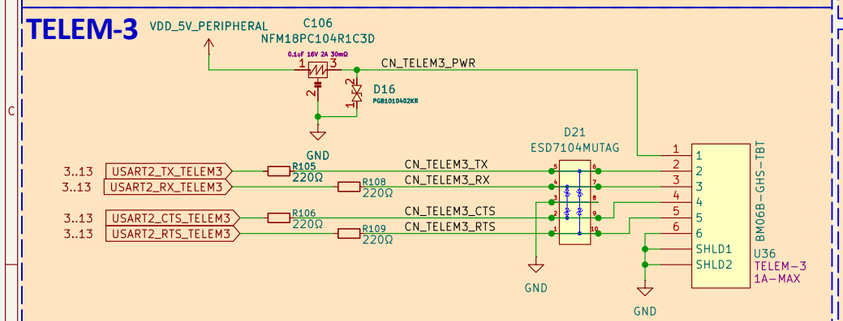
CAN-2



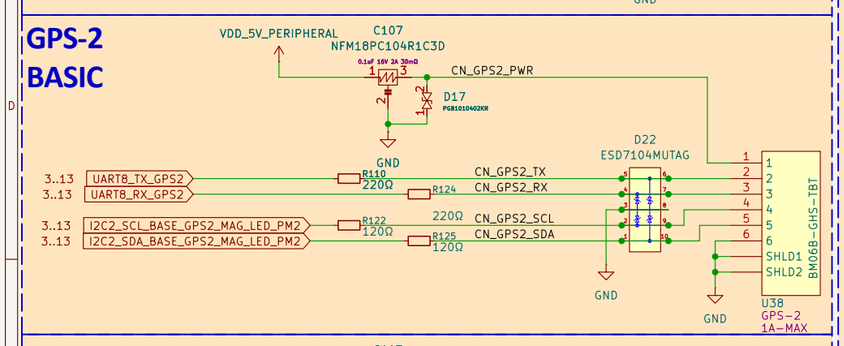
TELEM-2



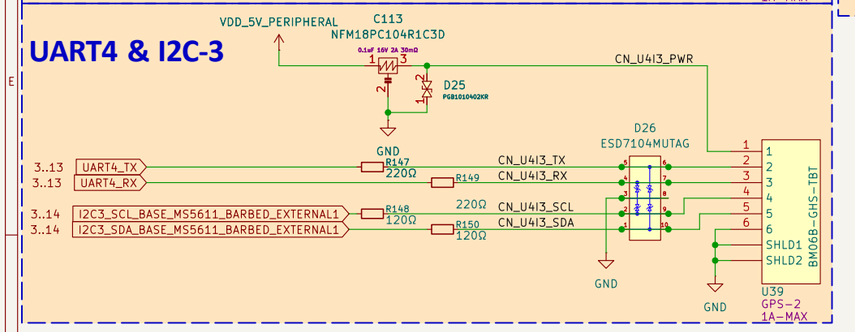
TELEM-3



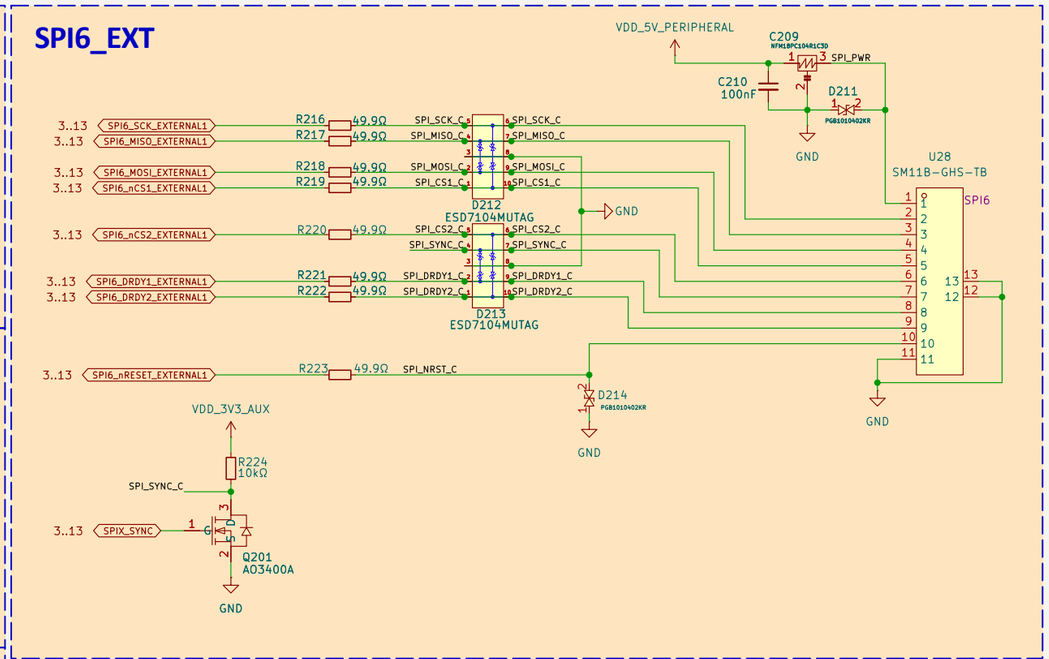
GPS-2 BASIC



UART4 & I2C-3

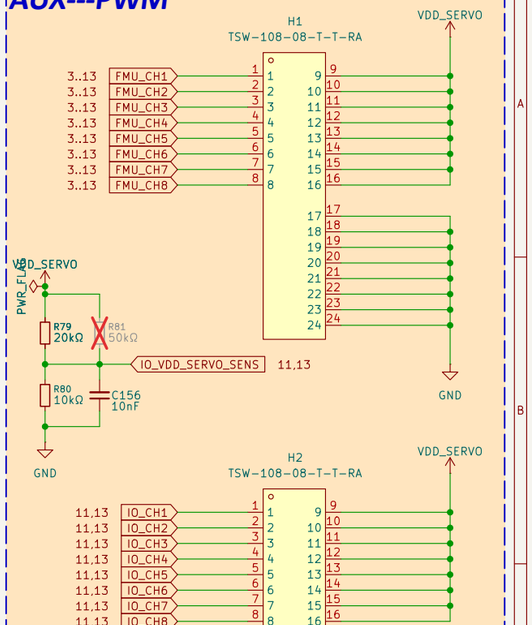


SPI6_EXT



I2C3

AUX---PWM

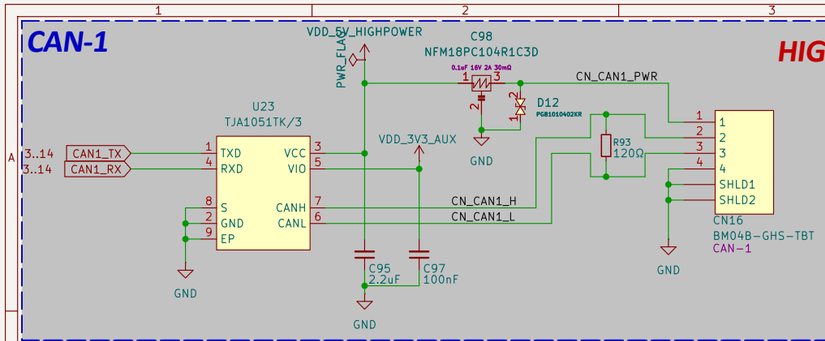


MAIN---PWM

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	**CARRIER_CONN_1**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 13 Total 14	
		REV0 VERO / A3 / VAP-SAN	

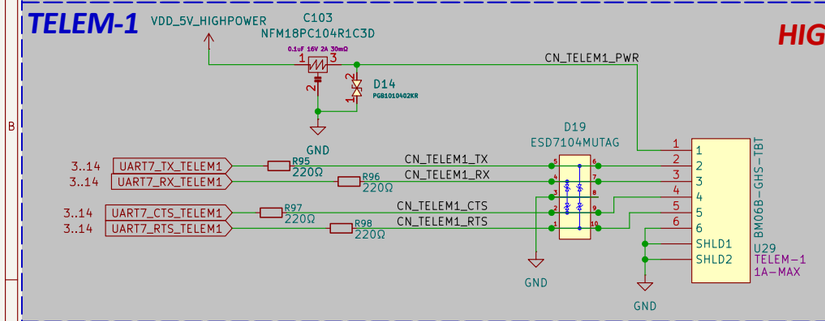
CAN-1

HIGH POWER



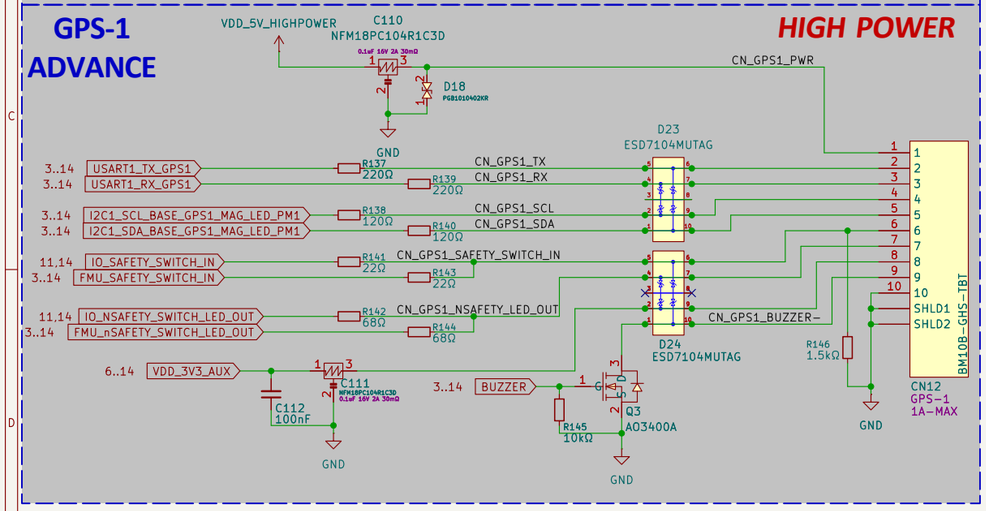
TELEM-1

HIGH POWER

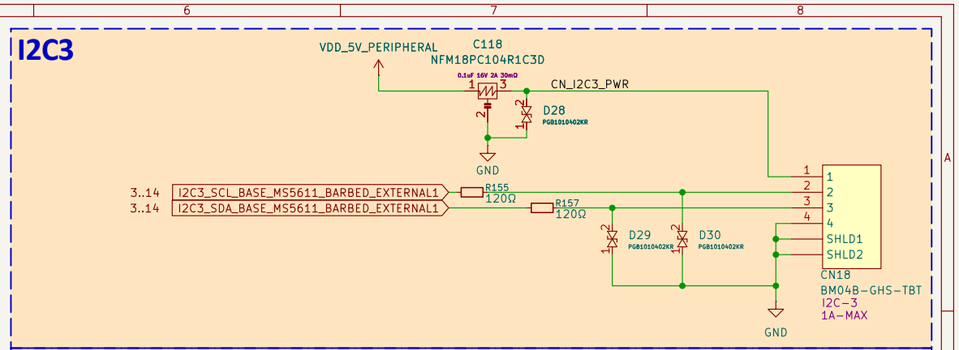


GPS-1 ADVANCE

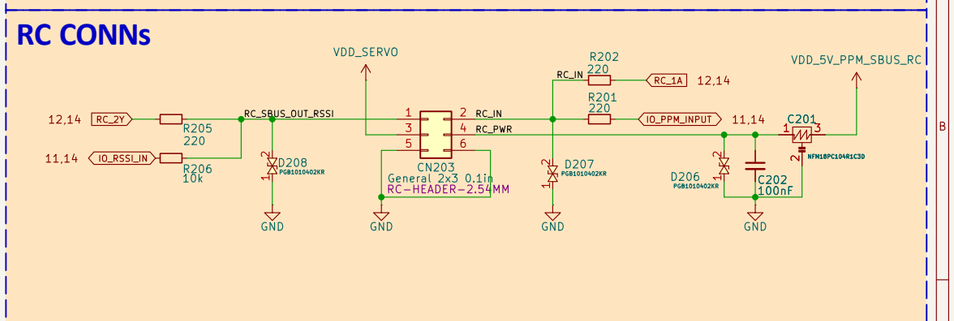
HIGH POWER



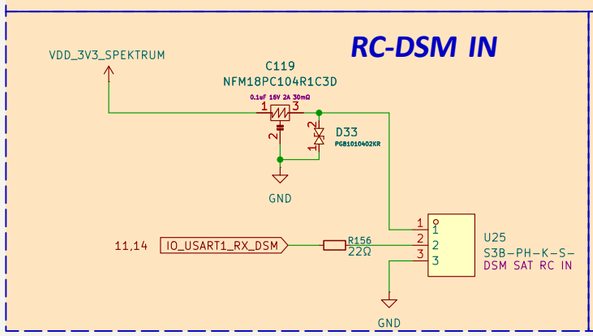
I2C3

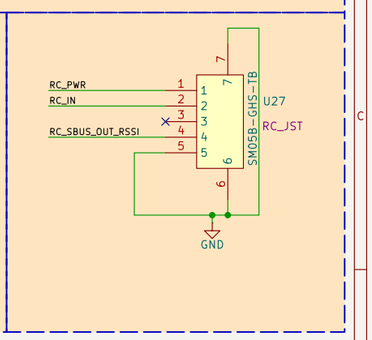


RC CONNs



RC-DSM IN





Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_CONN_2**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 14 Total 14
		REV0 VER0	A3	VAP-SAN